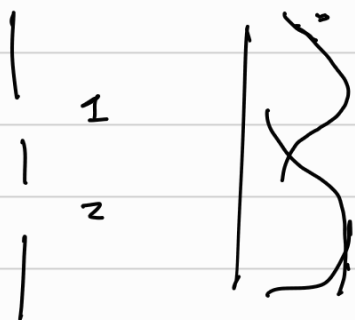


* Info is a fⁿ of probabilities: it is the entropy associated with the prob. distribution

$$I_{\text{info}} \sim K_{\text{info}} \ln(1/p)$$



$$P_c = P_1 + P_2$$

Classically

→ Ball going along a stream has
 $P_1 = .5, P_2 = .5$

• In QM: associate a Q prob. amp. w each slit ψ_1 and ψ_2

$$P_Q = |\psi_1 + \psi_2|^2$$

why? a state is given by
 $|\Phi\rangle = a_1 |\psi_1\rangle + a_2 |\psi_2\rangle$

which a superpositional sum of each Quantum state as such

$$P_Q = \langle \psi, \psi \rangle = \int d^4x \langle \psi | \psi \rangle$$

* $\langle \psi, \psi \rangle : L^2(\mathbb{R})$ norm

$\langle \psi | \psi \rangle$: "Density" norm which acts as the more familiar $\mathbb{R}^n$ -norm

$$\langle \psi | \psi \rangle = \psi^*(x) \psi(x)$$

→ This is why P is a probability density as $\langle \psi | \psi \rangle$.

Claim: $\psi = a_1 \psi_1 + a_2 \psi_2 \in L^2(\mathbb{R})$

$$\text{then } P_Q = |\psi|^2$$

$$= |\psi_1|^2 + |\psi_2|^2 + 2\sqrt{p_1 p_2} \cos(\theta)$$

Proof:

$$\langle \psi, \psi \rangle = a_1^* a_1 \langle \psi_1, \psi_1 \rangle + a_1^* a_2 \langle \psi_1, \psi_2 \rangle + a_2^* a_1 \langle \psi_2, \psi_1 \rangle + a_2^* a_2 \langle \psi_2, \psi_2 \rangle$$

→ Since ψ is an element of $L^2(\mathbb{R})$ then it is expanded in a complete orthonormal set wrt its eigenvalues/vectors as such

$$= a_1^* a_1 \int dx \psi_1^*(x) \psi(x) + \dots$$

$$= \int dx [|\psi_1|^2 + |\psi_2|^2 + \psi_1^* \psi_2 + \psi_2^* \psi_1]$$

→ Not too sure where the a_1 and a_2 went ... could only work with $a_1 = a_2 = 1 \dots$

$$* P_A \geq P_C \quad \text{or} \quad P_A \leq P_C$$

Conditional Prob:

• Consider a measurement with countably many outcomes

event: $A \leadsto$ Measurement

outcomes: $\{a_i\} \leadsto$ Possible measurements

$P(a_i) \equiv$ prob that a_i occurs $A=a_i$

$$0 \leq P(a_i) \leq 1$$

→ $\{a_i\}$: if it is a complete set of outcomes then

$$\sum_i P(a_i) = 1$$

* If we add a second event B w/ outcomes $\{b_j\}$ then $P(a_i)$ & $P(b_j)$ do not tell us everything

→ Joint probability: $\{P(a_i, b_j)\}$

$$P(a_i, b_j) \equiv P(a_i \wedge b_j)$$

, : comma denotes "and"

$$\left\{ \begin{array}{l} P(a_i, b_j) \equiv \text{prob. that } A = a_i \text{ AND} \\ \text{prob that } B = b_j \end{array} \right\}$$

* If A and B are independent then

$$P(a_i, b_j) = P(a_i)P(b_j)$$

why? Try cons:

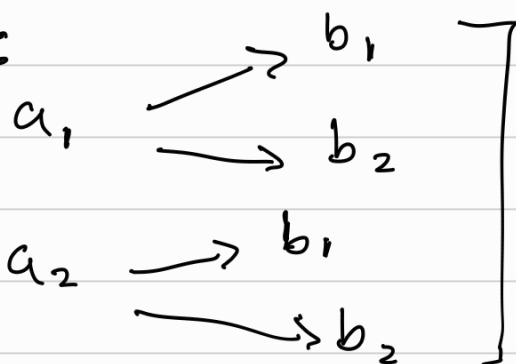
$$A: a_1 = .5, a_2 = .5$$

$$B: b_1 = .5, b_2 = .5$$

$$P(a_i) = .5$$

$$P(b_j) = .5$$

ex:



Thus, there are 4 possible outcomes for each with equal prob. of .25.

$$P(a_i, b_j) = .25$$

$$\text{As such this was } P(a_i, b_j) = P(a_i)P(b_j) = .5 \cdot .5 = .25$$

* How can I be more mathematical?

$$\langle \psi, B^\dagger \psi \rangle = \langle B \psi, \psi \rangle = \langle \psi, B \psi \rangle^*$$

$$\leadsto ? \quad \langle H \psi, H \psi \rangle = \langle \psi, \psi \rangle$$

$$L^2(\cdot) \Rightarrow \langle \psi, H H \psi \rangle - \langle \psi, \psi \rangle = 0$$

$$H^\dagger H \psi - \psi = (H^\dagger H - \mathbb{I}) \psi = 0$$

$$\langle \psi, \psi \rangle = \|\psi\|^2 \geq 0$$

$$\Rightarrow \langle \psi, \psi \rangle = 0 \quad \psi = 0$$

$$\Rightarrow H^p H - \mathbb{I} = 0$$